Supplemental Materials for: *Tree Type and Urban Growing Conditions Associated with Street Tree Stress: Lessons from Two U.S. Cities*

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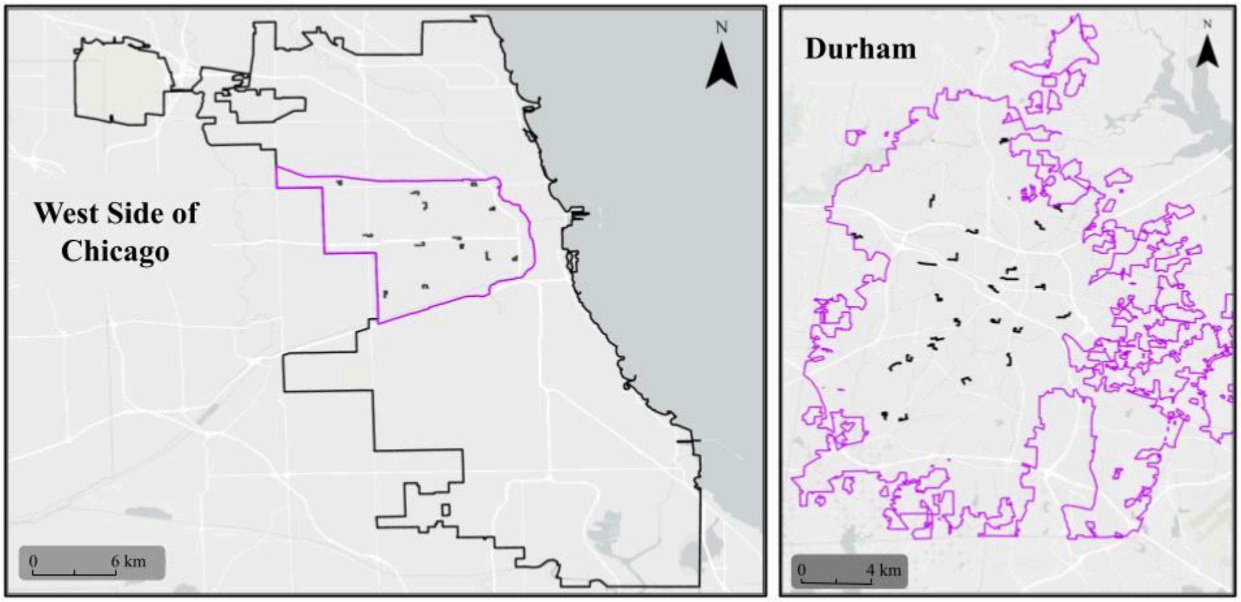

Figure 1: Sampling locations in Chicago, IL and Durham, NC. The small black lines each represent a sampled street segment while the purple outline shows the outline of the West Side of Chicago (City of Chicago, 2020) and the outline of the city (in Durham). The black outline in the Chicago map is the outline for the entire city of Chicago.

Table 1: Trees sampled each year in Chicago, IL and Durham, NC. In Chicago, trees resampled in 2023 were all previously sampled trees of 7 species that were common and represented both native and non-native species. These species were Elms (*Ulmus* species, including hybrid cultivars identified by tree tags), Norway maple (*Acer platanoides*), Freeman's maple (*Acer x freemanii*), Honeylocust (*Gleditsia triacanthos*), Kentucky coffeetree (*Gymnocladus dioicus*), Little-leaf linden (*Tilia cordata*), and Hackberry (*Celtis occidentalis*). In Durham, trees resampled in 2023 were all those accessible from 2022. Note that the column with "resampled from previous" indicates trees that were resampled from previous years, not those sampled as part of quality control. These numbers are slightly higher than the total number included because some samples had to be excluded due to incomplete data (e.g. no crown light data)

Year	Chicago total trees sampled	Chicago resampled from previous	Durham total trees sampled	Durham resampled from previous
2021	383	-	-	-
2022	390	105	273	-
2023	664	326	481	213

Table 2: Quality assessment results for street tree sampling in Durham and Chicago using categories in final analysis. A random sample of 10-25% of sampled trees were resampled within the same season to see if there is a difference between estimated tree health outcome variables and crown light between two repeat visits (measurement error). This table shows the results for categories that match those used in the analyses, which is a simplification of the raw categories. For example, the initial values for dieback ranged from 1 to 21, increasing in 5% intervals after the initial "trace" category. Since higher levels of dieback were rare, dieback was simplified in analyses down to only go from 1 to 6 instead of 1 to 21. Table 2 shows measurement error with the raw categories and table 2 in the main text shows the change from the raw categories to the categories used for analysis.

Dataset	# of Trees	Average Discoloration Difference	Average Defoliation Difference	Average Dieback Difference	Average Crown Stress Difference	Average Crown Light Exposure Difference
Durham 2022	27	0.407	0.074	0.370	0.370	0.333
Durham 2023	56	0.446	0.161	0.466	0.534	0.259
All Durham	83	0.434	0.133	0.435	0.482	0.282
Chicago 2021	23	0.435	0.435	0.478	0.435	0.391
Chicago 2022	69	0.435	0.319	0.565	0.754	0.385
Chicago 2023	71	0.324	0.155	0.310	0.366	0.225
All Chicago	163	0.387	0.264	0.442	0.540	0.314
All	246	0.402	0.220	0.440	0.520	0.303

Table 3: Quality assessment results for street tree sampling in Durham and Chicago using original categories recorded in the field. Note that there are no columns for crown stress and crown light exposure because those categories were not modified for analysis. See Table 1 in the main text for the adjustments to the tree health variable categories done for analysis and the corresponding initial values collected in the field.

Dataset	# of Trees	Average Discoloration Difference	Average Defoliation Difference	Average Dieback Difference
Durham 2022	27	0.407	0.074	0.593
Durham 2023	56	0.482	0.161	0.759
All Durham	83	0.458	0.133	0.706
Chicago 2021	23	0.435	0.435	1.000
Chicago 2022	69	0.449	0.319	0.855
Chicago 2023	71	0.324	0.155	0.465
All Chicago	163	0.393	0.264	0.706
All	246	0.415	0.220	0.706

Table 4: Category simplifications for explanatory variables used in data analysis. The "original category" column indicates the values for a given variable that were recorded in the field. The "new category" column indicates the new groupings used in analysis for a given variable based on variation present in the data. See supplementary Table 7 for the number of individual trees within each category.

City	**Variable**	**Original Category**	**New category**
Durham	Land Use Type	Single Family Residential - Attached	Residential
		Single Family Residential - Detached	
		Multi-family Residential	
		Commercial	Commercial/Mixed
		Mixed-use	
		Institutional	Institutional
		Vacant	Low intensity management
		Natural areas	
		Transportation	Other
		Agriculture	
	Site Type	Sidewalk Cutout	Restricted Rooting Space
		Planter box	
		Other Maintained Hardscape	
		Sidewalk Planting Strip	Sidewalk Planting Strip
		Median	Median
		Front yard	Yard
		Side yard	
		Other Maintained Area	Other Maintained
		Natural area	Natural area
Chicago	Land Use Type	Single Family Residential - Attached	Single Family Residential
		Single Family Residential - Detached	
		Multi-family Residential	Multi-family Residential
		Commercial	Commercial/Mixed use
		Mixed-use	
		Institutional	Institutional

		Managed Park	Park
		Agriculture	Other
		Vacant	
		Transportation	
		Industrial	
		Utility	
	Site Type	Sidewalk Cutout	Sidewalk Cutout
		Sidewalk Planting Strip	Sidewalk Planting Strip
		Other Maintained Area	Other
		Maintained Park	

Table 5: Species groups for trees in Chicago. Total number of observations is the total data points for a group while the number of trees is the number of unique trees in a group since some trees were resampled. Those in the "other" categories include species with fewer than 71 observations (5% of the total). Genera that, in aggregate, met the 71 observation threshold were included as their own group (excluding any individual species that separately met the 71 observation threshold). The "Other" group was split based on mature tree height (from Dirr (2016)), with small as ≤ 10.5m, medium between 10.5m and 15.25m, and large as > 15.25m. The [Base] identifies the baseline category in the model.

Group in Model	Number of Observations | Trees	Species Included (Observations | Trees)
Gleditsia triacanthos[Base]	299 | 159	*G. triacanthos* (299 | 159)
Acer planatoides	156 | 84	*A. platanoides* (156 | 84)
Celtis occidentalis	72 | 40	*C. occidentalis* (72 | 40)
Ulmus	186 | 111	*Ulmus* hybrid (136 | 87), *U. americana* (50 | 24)
Fraxinus	159 | 148	*F. americana, F. pennsylvanica, F. nigra* (grouped due to uncertain IDs)
Acer other	142 | 133	*A. griseum* (1 | 1), *A. palmatum* (2 | 2), *A. rubrum* (36 | 33), *A. saccharinum* (45| 45), *A. saccharum* (10 | 8), *A. x freemanii* (47| 43), Unknown (1 | 1)
Tilia	95 | 67	*T. americana* (48 | 40), *T. cordata* (46 | 26), *T. tomentosa* (1 | 1)
Other Large	147 | 114	*Aesculus flava* (4 | 2), *Aesculus hippocastanum* (1 | 1), *Betula nigra* (1 | 1), *Betula papyrifera* (1 | 1), *GInkgo biloba* (21 | 17), *Gymnocladus dioicus* (63 | 37), *Liquidambar styraciflua* (3 | 3), *Metasequoia glyptostroboides* (1 | 1), *Picea unknown* (4 | 3), *Pinus unknown* (4 | 4), *Platanus hybrida* (8| 7), *Platanus occidentalis* (3 | 3), *Populus deltoides* (2 | 2), *Quercus alba* (1 | 1), *Quercus bicolor* (11 | 11), *Quercus imbricaria* (9 | 9), *Quercus macrocarpa* (4 | 4), *Quercus rubra* (2 | 2), *Taxodium distichum* (4 | 4)
Other Medium	107 | 94	*Aesculus unknown* (1 |1), *Ailanthus altissima* (19 | 18), *Alnus glutinosa* (2 | 1), *Catalpa speciosa* (19 | 16), *Diospyros virginiana* (1 | 1), *Juniperus virginiana* (1 | 1), *Morus alba* (10 | 8), *Morus rubra* (2 | 1), *Nyssa silvatica* (3 | 3), *Picea pungens* (2 | 2), *Populus tremuloides* (4 | 4), *Pyrus calleryana* (14 | 11), *Quercus muhlenbergii* (13 |13), *Robinia pseudoacacia* (9 | 9), *Taxus unknown* (1 | 1), *Thuja occidentalis* (6 | 4)
Other Small	61 | 48	*Aesculus glabra* (2 | 2), *Amelanchier x grandiflora* (1 | 1), *Cercis canadensis* (4 | 3) , *Crataegus crus-galli* (3 | 3), *Magnolia liliiflora* (2 | 1), *Malus unknown* (9 | 7), *Prunus cerasifera* (15 | 9), *Prunus serrulata* (8 | 6), *Prunus unknown* (3 | 2), *Syringa reticulata* (3 | 3), *Thuja unknown* (11 | 11)

Table 6: Species groups for trees in Durham. Total number of observations is the total data points available for a given genus (or species) while the number of trees describes the number of unique trees in a group since some trees were sampled more than once. Those in the "other" categories include species with fewer than 41 observations (5% of the total). Genera that, in aggregate, met the 41 observation threshold were included as their own group (excluding any individual species that separately met the 41 observation threshold). The "Other" group was split based on mature tree size (from Dirr (2016)), with small as ≤ 10.5m, medium between 10.5m and 15.25m, and large as > 15.25m. The [Base] identifies the baseline category in the model.

Group in Model	Number of Observations | Trees	Species Included (Observations | Trees)
Acer rubrum	67 | 42	*A. rubrum* (67 | 42)
Cercis canadensis	58 | 35	*Cercis canadensis* (58 | 35)
Ulmus americana	48 | 32	*Ulmus americana* (48 | 32)
Quercus phellos	43 | 36	*Quercus phellos* (43 | 36)
Lagerstroemia[Base]	83 | 69	*L. indica* (13 | 13), Unknown (70 | 56)
Quercus other	50 | 39	*Q. bicolor* (10 | 10), *Q. coccinea* (6 | 3), *Q. lyrata* (7 | 6), *Q. macrocarpa* (6 | 5), *Q. rubra* (1 | 1), *Q. virginiana* (15 | 11), Unknown (5 | 3)
Prunus	49 | 37	*P. persica* (4 | 4), *P. serotina* (3 | 3), *P. yedoensis* (16 | 13), Unknown (26 | 17)
Acer other	47 | 28	*A. buergerianum* (14 | 9), *A. campestre* (2 | 1), *A. palmatum* (3 | 3), *A. platanoides* (11 | 6), *A. platanoides x truncatum* (2 | 1), *A. saccharum* (8 | 5), *A. truncatum* (6 | 3), *A. x freemanii* (2 | 1), Unknown (1 | 1)
Other large	136 | 100	*Betula nigra* (2 | 1), *Carya illinoinensis* (3 | 3), *Ginkgo biloba* (17 | 11), *Gymnocladus. dioicus* (26 | 13), *Liquidambar styraciflua* (27 | 27), *Liriodendron tulipifera* (2 | 2), *Magnolia grandiflora* (2 | 2), *Morus rubra* (1 | 1), *Pinus taeda* (5 | 5), *Platanus acerifolia* (1 | 1), *Platanus occidentalis* (1 | 1), *Taxodium ascendens* (3 | 3), *Taxodium distichum* (4 | 4), *Tilia cordata* (2 | 1), *Ulmus alata* (11 | 6), *Zelkova serrata* (30 | 20)
Other medium	52 | 35	*Cladrastis kentukea* (1 | 1), Ilex unknown (1 | 1), *Juniperus virginiana* (7 | 5), *Nyssa sylvatica* (1 | 1), *Pyrus calleryana* (2 | 2), *Ulmus parvifolia* (10 | 5), *Ulmus unknown* (8 | 4)
Other small	121 | 88	*Albizia julibrissin* (1 | 1), *Amelanchier canadensis* (1 | 1), *Amelanchier* unknown (1 | 1), *Amelanchier x grandiflora* (2 | 1), *Asimina triloba* (5 | 3), *Carpinus caroliniana* (28 | 14), *Carpinus* unknown (1 | 1), *Cercis reniformis* (4 | 4), *Chionanthus retusus* (12 | 12), *Cornus florida* (6 | 3), *Cornus kousa* (3 | 2), *Crataegus viridis* (8 | 4), *Ficus carica* (2 | 2), *Ilex cornuta* (1 | 1), *Magnolia stellata* (9

		| 9), *Magnolia virginiana* (10 | 5), *Melia azedarach* (2 | 1), *Parrotia persica* (4 | 2), *Pistacia chinensis* (20 | 20), *Pyrus communis* (1 | 1)

Table 7: Number of trees with each for variables used in growing condition models.

Variable	Type	Durham	Chicago
Number of Trees	Total number of unique trees across all years	541	998
Land Use	Single-family residential	322	515
	Multi-family residential		323
	Commercial or Mixed	97	96
	Institutional	89	17
	Park	0	26
	Low-management (Natural area, Vacant)	15	0
	Other	18	21
Percent Impervious Surface	1 (< 25%)	184	26
	2 (26-50%)	135	44
	3 (51-75%)	118	248
	4 (76-100%)	101	680
Power line	Present	96	244
	Absent	445	754
Mulched	According to BMP	167	299
	Not according to BMP	54	42
	Not mulched	320	657
Pruned	According to BMP	254	683
	Not according to BMP	112	246
	Not pruned	175	69
Landscaping Intensity	1 (minimal or no landscaping)	119	152
	2 (evidence of mowing or minor landscaping)	314	627
	3 (landscaping with focus on the tree)	57	123
	4 (landscaping throughout plantable area)	51	96
Site Type	Most Restrictive (Sidewalk cutout, Planter box, Other Hardscape)	61	188
	Sidewalk planting strip	229	783
	Yard (front, side, or back)	140	-
	Median	58	-

	Other Managed	53	-
	Other	0	27

Table 8: Description of tree stress variables measured as part of the US Forest Service and The Nature Conservancy Healthy Trees, Healthy Cities protocols (Hallett et al., 2019). The last column describes how these categories were simplified for analysis based on the variability present within our sample (i.e. few trees had high dieback, so the higher categories were collapsed). Quality controls were done by replicating data collection for a subset of trees, within a given year, results available in supplementary Tables 1 and 2)

Tree stress variable	**Description**	**Categories**	**Simplified categories**
Leaf discoloration	Percent of total leaf area that is discolored from the normal range of color for the species/cultivar	(1) trace (<2%)	(1) trace (<2%)
		(2) 2-25%	(2) 2-25%
		(3) 26-50%	(3) 26-50%
		(4) 51-75% (5) 76-100%	(4) 51%+
Leaf defoliation	Percent of total leaf area that is missing due to holes or missing sections in the leaves. Note: this requires that some part of the leaf remains.	(1) trace (<2%)	(1) trace (<2%)
		(2) 2-25%	(2) 2-25%
		(3) 26-50% (4) 51-75% (5) 76-100%	(3) 26%+
Fine twig dieback	Percent of total leaf area that is missing due to lost leaves. Only count leaves on small twigs on the outer edge of the tree canopy.	(1) trace (<2%)	(1) trace (<2%)
		(2) 2-5%	(2) 2-5%
		(3) 6-10%,	(3) 6-10%
		(4) 11-15% (5) 16-20% (6) 21-25%	(4) 11-25%
		(7) 26-30% (8) 31-35% (9) 36-40% (10) 41-45% (11) 46-50% (12) 51-55% (13) 56-60% (14) 61-65% (15) 66-70% (16) 71-75% (17) 76-80% (18) 81-85% (19) 86-90% (20) 91-95% (21) 96-100%	(5) 26%+
Crown stress	An overall measure of tree stress that incorporates the three measures above alongside large branch dieback.	1 (< 10% cumulative dieback, discoloration, and defoliation and no major branch mortality)	Same as original
		2 (10-25% cumulative dieback, discoloration, and defoliation and/or 25% or less crown area missing due to broken or dead large branches)	
		3 (26-50% cumulative dieback, discoloration, and defoliation and/or 50% or less crown area missing due to broken or dead large branches)	
		4 (more than 50% cumulative dieback, discoloration, and defoliation and/or more than 50% crown area missing due to broken or dead large branches)	
		5 (dead)	

Table 9: Predictors removed from models in Chicago and Durham. VIF stands for variable inflation factors from running versions of the models with the predictors included.

City	Predictor	Reason
Both	Nearby street	Not enough variation in Chicago
Both	Nearby sidewalk	Not enough variation in Chicago
Both	Tree guards	Not enough trees with guards in Chicago
Both	Tree gator bags, staking	Rare in both cities
Both	Percent Impervious Surface	VIF > 5
Both	Crown Light	VIF > 5

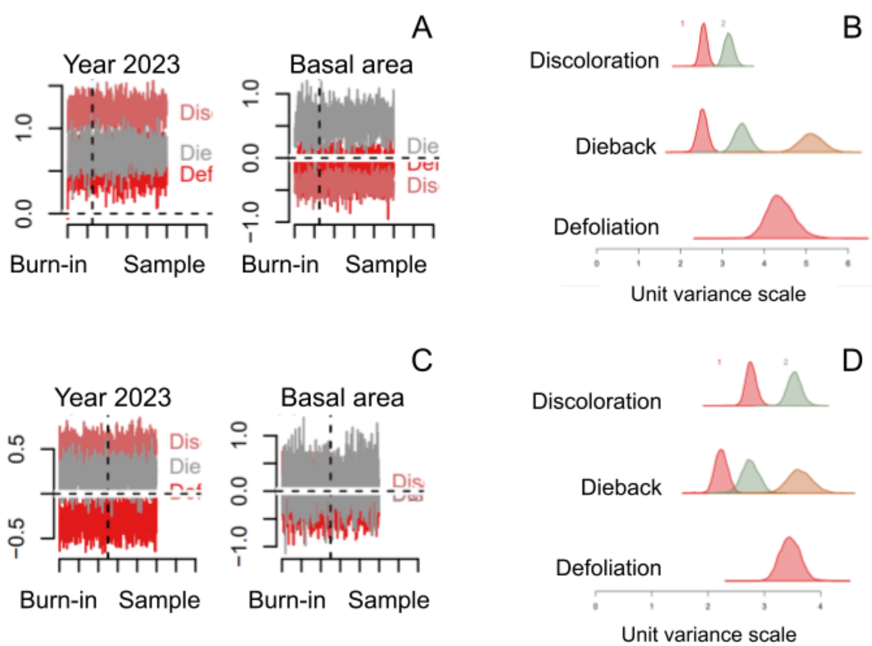

Figure 2: Model diagnostics for best fitting multivariate models in Chicago (A,B) and Durham (C,D). The multivariate models have Discoloration, Defoliation, and Dieback as ordinal response variables. Panels A, C show example chains for 2023 (compared to 2021 in Chicago and 2022 in Durham) and for basal area parameter estimates. The vertical dashed line shows where the burn-in ends and the samples used to estimate the posterior begin. Panels B, D show the partition estimates for the three ordinal response variables with discoloration having three levels, Dieback having four, and Defoliation having two. For both cities, the burn-in was set to 5000 and in Durham the model was run for 10,000 iterations whereas for Chicago (with more trees and years) it was run for 20,000 iterations to ensure the posterior was well sampled.

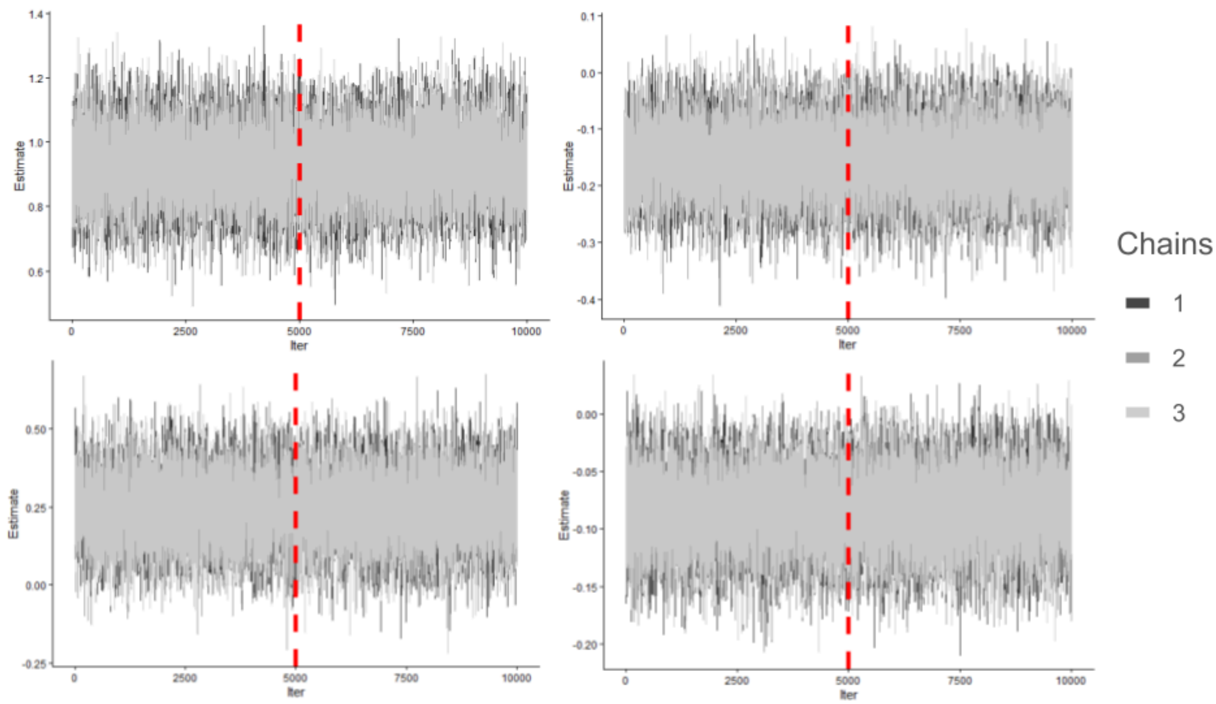

Figure 3: Chains for best fitting univariate (crown stress) models in Chicago (A,B) and Durham (C,D). As examples of chain convergence, panels A, C show example chains for 2023 (compared to 2021 in Chicago and 2022 in Durham) and panels B, D show the chains for basal area. The vertical dashed line shows where the burn-in ends and the samples used to estimate the posterior begin. For both cities, the burn-in was set to 5000 (red dashed line) and the model was run for 10,000 iterations. The models were run with three chains, colored in greyscale.

Table 10: DIC values for models tested for multivariate model in Chicago. Response variables are defoliation, discoloration, and dieback. The variables noted in table 7 were already excluded due to lack of variation or high variable inflation factors. The bolded row is the model with the lowest DIC, used for interpretation in the main text.

Model Formula	DIC
(Base model) Response ~ Segment + Year + Basal Area x Species Group + Powerline + Mulch + Landscaping + Pruning + Urban Tolerant x Site Type + Land Use	**42118.89**
Base model without pruning	42482.05
Base model without powerline	42458.9
Base model without landscaping	42359.97
Base model without basal area x species group interaction (just additive)	42362.25
Base model without urban tolerant x site type interaction (just additive)	42374.87

Table 11: DIC values for models tested for univariate model in Chicago. The response variable is crown stress. The variables noted in table 7 were already excluded due to lack of variation or high variable inflation factors. The bolded row is the model with the lowest DIC, used for interpretation in main text.

Model Formula	**DIC**
(Base model) Response ~ Segment + Year + Basal Area x Species Group + Powerline + Mulch + Landscaping + Pruning + Urban Tolerant x Site Type + Land Use	2765.5
Base model without powerline	2762.6
Base model without powerline or landscaping	2756.3
Base model without powerline, landscaping, or land use	2751.7
Base model without powerline, landscaping, or pruning	2756.1
Base model without powerline, landscaping, pruning, or land use	2753
Base model without powerline, landscaping, land use, or mulch	2751.1
Base model without powerline, landscaping, land use, mulch, or pruning	**2750.2**
Base model without powerline, landscaping, land use, mulch, pruning, or basal area x species group interaction (just additive)	2763.5
Base model without powerline, landscaping, land use, mulch, pruning, or urban tolerant x site type interaction (just additive)	2753.3

Table 12: DIC values for models tested for multivariate model in Durham. Response variables are defoliation, discoloration, and dieback. The variables noted in table 7 were already excluded due to lack of variation or high variable inflation factors. The bolded row is the model with the lowest DIC, used for interpretation in the main text.

Model Formula	DIC
(Base model) Response ~ Segment + Year + BasalArea*SppGroup + Powerline + Mulch + Gardenscape + UrbanApp*SiteType + LandUse	23923.38
Base model without powerline	**23825.96**
Base model without powerline and mulch	23887.71
Base model without powerline and urban tolerance	24391.8

Table 13: DIC values for models tested for univariate model in Durham. The response variable is crown stress. The variables noted in table 7 were already excluded due to lack of variation or high variable inflation factors. The bolded row is the model with the lowest DIC, used for interpretation in main text.

Model Formula	DIC
(Base model) Response ~ Segment + Year + BasalArea x SppGroup + Powerline + Mulch + Gardenscape + PrunedCorrect + UrbanApp x SiteType + LandUse	1850.8
Base model without basal area x species group (just additive)	1839.6
Base model without pruning or basal area x species group (just additive)	**1837.7**
Base model without pruning or either interaction (basal area x species group and urban tolerance x site type)	1842.1
Base model without pruning, mulch, or basal area x species group (just additive)	1837.8
Base model without pruning, mulch, urban tolerance, or basal area x species group (just additive)	1855.3

Table 14: All predictors tested and those present in the best-fitting models in Chicago and Durham (assessed through DIC). Note that some predictors were not used at all due to high VIF values or limited variation (see Table 9). If a predictor is present in the best-fitting model, the corresponding box is marked with an x. Each city has a multivariate ordinal model (with discoloration, defoliation, and dieback all as response variables) and a univariate model (with just crown stress as a response variable).

Predictor	Chicago		Durham	
	Multivariate: Discoloration, Defoliation, Dieback	Univariate: Crown Stress	Multivariate: Discoloration, Defoliation, Dieback	Univariate: Crown Stress
Street Segment	x	x	x	x
Year	x	x	x	x
Land Use	x		x	x
Site Type	x	x	x	x
Crown Light	Excluded due to high VIF			
Power lines	x			x
Percent Impervious Surface	Excluded due to high VIF			
Landscaping	x		x	x
Mulch	x		x	x
Pruning	x			
Species Group	x	x	x	x
Urban Tolerance	x	x	x	x
Basal Area	x	x	x	x
Basal Area x Species Group	x	x	x	
Urban Tolerant x Site Type	x	x	x	x

Table 15: All parameter estimates for multivariate (discoloration, defoliation, dieback) growing condition model in Chicago. The baseline for street segment is the wealthiest street segment (arbitrarily numbered 1), the baseline for year is 2021, the baseline for species group is the most abundant species (*Gleditsia triacanthos*), the baseline for landscaping intensity is no landscaping, the baseline for site type is the most restrictive rooting zone (sidewalk cutout), and the baseline for land use is single-family residential. SE is the standard error, CI_025 and CI_975 together describe the 95% confidence interval around the parameter estimate. Sig95 indicates whether or not the 95% confidence interval crosses zero.

Response	Predictor	Predictor Level	Estimate	SE	CI_025	CI_975	sig95
Defoliation	Segment	2	0.0672	0.482	-0.863	1.01	
		3	-0.379	0.424	-1.23	0.437	
		4	-0.492	0.415	-1.33	0.296	
		5	-0.66	0.449	-1.56	0.2	
		6	-0.0198	0.375	-0.754	0.726	
		7	-1.45	0.524	-2.53	-0.479	*
		8	-0.724	0.537	-1.8	0.293	
		9	-0.244	0.421	-1.08	0.56	
		10	-0.236	0.359	-0.946	0.471	
		11	0.623	0.323	0.00213	1.27	*
		12	-0.798	0.482	-1.75	0.153	
		13	-0.462	0.469	-1.4	0.427	
	Year	2022	0.724	0.204	0.328	1.13	*
		2023	0.585	0.189	0.22	0.959	*
	Basal Area		-0.114	0.131	-0.371	0.139	
	Species Group	*Acer platanoides*	2.12	0.922	0.383	3.98	*
		Celtis occidentalis	3.2	0.895	1.52	4.99	*
		Fraxinus	-0.476	0.843	-2.12	1.18	
		Tilia	0.578	0.682	-0.796	1.91	
		Ulmus	4.77	0.693	3.5	6.2	*
		Other large	1.79	0.706	0.425	3.2	*
		Other medium	0.33	0.665	-0.991	1.61	
		Other small	0.123	1.06	-2.06	2.17	
	Powerline		0.00112	0.203	-0.408	0.388	

	Mulch	Correct	0.0854	0.212	-0.335	0.497	
		Volcano	-0.0893	0.398	-0.904	0.66	
	Landscaping	Minor	0.0278	0.22	-0.401	0.469	
		Moderate	-0.228	0.324	-0.885	0.384	
		Extensive	-0.26	0.351	-0.972	0.41	
	Pruned following BMPs		0.0585	0.167	-0.261	0.395	
	Urban Tolerant		0.0324	0.458	-0.862	0.924	
	Site Type	Other	0.0154	0.979	-1.97	1.89	
		Sidewalk Planting Strip	0.0913	0.277	-0.452	0.639	
	Land Use	Commercial and Mixed	-0.0832	0.343	-0.75	0.584	
		Institutional	-0.234	0.692	-1.6	1.1	
		Multi-family Residential	0.129	0.182	-0.229	0.486	
		Managed Park	-1.49	0.676	-2.84	-0.188	*
		Other	0.591	0.485	-0.389	1.52	
	Basal Area x Species Group	*Acer platanoides*	0.622	0.323	0.0303	1.3	*
		Celtis occidentalis	-0.0294	0.233	-0.487	0.42	
		Fraxinus	0.0486	0.3	-0.529	0.65	
		Tilia	-0.0787	0.206	-0.493	0.315	
		Ulmus	0.233	0.177	-0.113	0.58	
		Other large	0.252	0.171	-0.0827	0.591	
		Other medium	-0.0047	0.172	-0.342	0.337	
		Other small	-0.0626	0.232	-0.53	0.38	
	Urban Tolerance x Site Type	Other	1.47	1.56	-1.61	4.49	
		Sidewalk Planting Strip	-0.655	0.432	-1.53	0.16	
Discoloration	Segment	2	0.967	0.32	0.335	1.59	*
		3	0.649	0.257	0.143	1.16	*
		4	0.548	0.234	0.0938	1.01	*

		5	0.375	0.247	-0.103	0.861	
		6	0.806	0.243	0.328	1.28	*
		7	0.203	0.247	-0.279	0.68	
		8	0.378	0.294	-0.198	0.958	
		9	0.0121	0.263	-0.512	0.512	
		10	0.32	0.224	-0.12	0.755	
		11	0.615	0.209	0.207	1.03	*
		12	0.689	0.293	0.116	1.27	*
		13	-0.186	0.266	-0.709	0.329	
	Year	2022	0.481	0.136	0.216	0.746	*
		2023	1.2	0.128	0.952	1.45	*
	Basal Area		-0.26	0.0775	-0.414	-0.109	*
	Species Group	*Acer platanoides*	0.694	0.456	-0.182	1.59	
		Celtis occidentalis	0.46	0.557	-0.643	1.55	
		Fraxinus	-0.689	0.397	-1.46	0.081	
		Tilia	0.423	0.43	-0.41	1.26	
		Ulmus	-0.578	0.378	-1.32	0.15	
		Other large	1.06	0.414	0.243	1.88	*
		Other medium	-0.0382	0.373	-0.775	0.696	
		Other small	-0.686	0.696	-2.07	0.664	
	Powerline		-0.0952	0.12	-0.336	0.138	
	Mulch	Correct	0.139	0.123	-0.0982	0.378	
		Volcano	-0.0586	0.248	-0.541	0.428	
	Landscaping	Minor	0.171	0.131	-0.0833	0.43	
		Moderate	-0.00628	0.178	-0.36	0.34	
		Extensive	0.131	0.198	-0.265	0.517	
	Pruned following BMPs		-0.0806	0.099	-0.274	0.113	
	Urban Tolerant		-0.348	0.27	-0.877	0.18	
	Site type	Other	0.238	0.523	-0.782	1.28	

		Sidewalk Planting Strip	-0.199	0.166	-0.521	0.126	
	Land Use	Commercial and Mixed	0.477	0.203	0.076	0.879	*
		Institutional	0.199	0.408	-0.605	0.987	
		Multi-family Residential	-0.0829	0.107	-0.293	0.126	
		Managed Park	0.085	0.321	-0.542	0.711	
		Other	0.155	0.302	-0.436	0.74	
	Basal Area x Species Group	*Acer platanoides*	0.417	0.144	0.139	0.704	*
		Celtis occidentalis	0.265	0.155	-0.0395	0.572	
		Fraxinus	0.107	0.143	-0.171	0.388	
		Tilia	0.49	0.142	0.215	0.772	*
		Ulmus	0.203	0.119	-0.0307	0.437	
		Other large	0.424	0.104	0.221	0.631	*
		Other medium	0.213	0.102	0.0131	0.414	*
		Other small	0.0985	0.154	-0.202	0.403	
	Urban Tolerance x Site Type	Other	-1.29	0.904	-3.08	0.447	
		Sidewalk Planting Strip	0.161	0.254	-0.331	0.661	
Dieback	Segment	2	2.14	0.609	0.942	3.35	*
		3	0.114	0.442	-0.728	1	
		4	-0.116	0.398	-0.907	0.67	
		5	1.18	0.437	0.324	2.05	*
		6	-0.279	0.42	-1.09	0.541	
		7	-0.427	0.41	-1.24	0.371	
		8	1.17	0.539	0.0977	2.25	*
		9	-0.181	0.435	-1.03	0.673	
		10	0.474	0.37	-0.248	1.22	
		11	-0.167	0.37	-0.897	0.565	
		12	-1.34	0.556	-2.45	-0.271	*

		13	0.525	0.443	-0.336	1.41	
	Year	2022	0.3	0.155	-0.00542	0.604	
		2023	0.703	0.148	0.414	0.993	*
	Basal Area		0.364	0.134	0.0992	0.626	*
	Species Group	*Acer platanoides*	1.28	0.863	-0.4	3	
		Celtis occidentalis	-0.867	0.991	-2.77	1.1	
		Fraxinus	2.02	0.686	0.686	3.38	*
		Tilia	-2.63	0.798	-4.21	-1.05	*
		Ulmus	0.857	0.66	-0.42	2.16	
		Other large	0.0894	0.714	-1.3	1.47	
		Other medium	0.697	0.609	-0.507	1.89	
		Other small	-0.84	1.27	-3.35	1.65	
	Powerline		-0.0714	0.203	-0.474	0.334	
	Mulch	Correct	0.264	0.225	-0.177	0.704	
		Volcano	0.856	0.405	0.0626	1.65	*
	Landscaping	Minor	-0.37	0.241	-0.836	0.104	
		Moderate	-0.224	0.314	-0.84	0.395	
		Extensive	-0.431	0.362	-1.15	0.264	
	Pruned following BMPs		0.017	0.181	-0.339	0.372	
	Urban Tolerant		-1.68	0.483	-2.61	-0.723	*
	Site type	Other	0.357	0.895	-1.37	2.12	
		Sidewalk Planting Strip	-0.828	0.3	-1.42	-0.247	*
	Land Use	Commercial and Mixed	0.838	0.371	0.108	1.57	*
		Institutional	0.879	0.724	-0.526	2.33	
		Multi-family Residential	0.0514	0.165	-0.268	0.377	
		Managed Park	-0.785	0.581	-1.94	0.347	
		Other	-1.32	0.519	-2.35	-0.31	*

	Basal Area x Species Group	*Acer platanoides*	0.195	0.286	-0.37	0.763	
		Celtis occidentalis	-0.651	0.282	-1.21	-0.102	*
		Fraxinus	0.615	0.267	0.0955	1.14	*
		Tilia	-0.509	0.26	-1.02	0.00654	
		Ulmus	0.329	0.216	-0.0865	0.765	
		Other large	-0.173	0.182	-0.529	0.183	
		Other medium	-0.186	0.179	-0.539	0.166	
		Other small	-0.339	0.286	-0.904	0.222	
	Urban Tolerance x Site Type	Other	0.543	1.54	-2.47	3.58	
		Sidewalk Planting Strip	0.811	0.461	-0.103	1.7	

Table 16: All parameter estimates for univariate (crown stress) growing condition model in Chicago. The baseline for street segment is the wealthiest street segment (arbitrarily numbered 1), the baseline for year is 2021, the baseline for Genus is all the groups with fewer than 40 observations (see Appendix B Table 3), the baseline for landscaping intensity is no landscaping, and the baseline for land use is single-family residential. SE is the standard error, CI_025 and CI_975 together describe the 95% confidence interval around the parameter estimate. Sig95 indicates whether or not the 95% confidence interval crosses zero.

Predictor	Predictor Level	Estimate	SE	CI_025	CI_975	sig95
(Intercept)		-0.3938	0.2714	-0.9278	0.1261	
Segment	2	0.587	0.2278	0.1402	1.032	*
	3	0.494	0.1878	0.1286	0.8657	*
	4	0.3328	0.1864	-0.03454	0.7017	
	5	0.75	0.1831	0.3936	1.111	*
	6	0.6178	0.1848	0.2605	0.987	*
	7	0.1588	0.191	-0.2196	0.5274	
	8	0.6039	0.2133	0.194	1.025	*
	9	0.05101	0.2149	-0.3671	0.4707	
	10	0.1134	0.1815	-0.24	0.4705	
	11	0.5277	0.1539	0.2276	0.8296	*
	12	0.009728	0.2254	-0.4282	0.4497	
	13	0.2066	0.192	-0.1729	0.5803	
Year	2022	0.5186	0.1218	0.2794	0.7587	*
	2023	0.9346	0.108	0.7218	1.146	*
BasalArea		-0.1543	0.06061	-0.2738	-0.03468	*
Species Group	*Acer platanoides*	0.5395	0.369	-0.1906	1.267	
	Celtis occidentalis	0.09568	0.4336	-0.7481	0.9408	
	Fraxinus	0.3688	0.322	-0.2676	1.005	
	Tilia	-0.2954	0.3376	-0.9531	0.3638	
	Ulmus	0.6766	0.2863	0.1131	1.245	*
	Other large	0.4327	0.3395	-0.2381	1.092	
	Other medium	0.1989	0.2931	-0.372	0.7696	
	Other small	-1.188	0.5698	-2.304	-0.0724	*
Urban Tolerance		-0.4557	0.2186	-0.8943	-0.03202	*
Site Type	Other	0.3768	0.419	-0.4432	1.195	

	Sidewalk Planting Strip	-0.3095	0.131	-0.5651	-0.05066	*
Basal area x Species Group	*Acer platanoides*	0.3303	0.1151	0.1063	0.5611	*
	Celtis occidentalis	0.06325	0.1217	-0.1756	0.3001	
	Fraxinus	0.3401	0.1211	0.1035	0.5781	*
	Tilia	0.1755	0.106	-0.02933	0.3844	
	Ulmus	0.3628	0.09055	0.1857	0.5395	*
	Other large	0.2244	0.08389	0.05809	0.3858	*
	Other medium	0.1616	0.08167	0.001257	0.3216	*
	Other small	-0.1003	0.1258	-0.346	0.1446	
Urban tolerance x Site type	Other	-1.36	0.7066	-2.762	0.0273	
	Sidewalk Planting Strip	0.2309	0.2004	-0.1663	0.6277	

Table 17: All parameter estimates for multivariate (discoloration, defoliation, dieback) growing condition model in Durham. The baseline for street segment is the wealthiest street segment with at least 10 trees (arbitrarily numbered 1), the baseline for year is 2022, the baseline for Species Group is the most common group (*Lagerstroemia*), the baseline for landscaping intensity is no landscaping, and the baseline for land use is residential. SE is the standard error, CI_025 and CI_975 together describe the 95% confidence interval around the parameter estimate. Sig95 indicates whether or not the 95% confidence interval crosses zero.

Response	Predictor	Predictor Level	Estimate	SE	CI_025	CI_975	sig95
Defoliation	intercept		-1.53	0.843	-3.21	0.107	
	Segment	2	1.95	0.741	0.587	3.47	*
		3	1.6	0.842	0.0375	3.25	*
		4	1.44	0.811	-0.166	3.06	
		5	0.235	0.88	-1.49	1.97	
		6	-0.758	0.758	-2.27	0.741	
		7	0.823	0.739	-0.639	2.31	
		Other	0.875	0.722	-0.517	2.27	
		8	0.217	0.758	-1.28	1.71	
		9	0.359	0.823	-1.25	1.98	
		10	1.06	0.749	-0.427	2.57	
		11	0.564	0.831	-1.11	2.19	
		12	1.07	0.737	-0.277	2.59	
		13	1.49	0.786	-0.0781	3.02	
		14	1.01	0.744	-0.44	2.46	
	Year	2023	-0.295	0.144	-0.577	-0.0128	*
	Basal Area		-0.0964	0.0977	-0.29	0.0929	
	Species Group	*Acer rubrum*	-2.82	0.818	-4.47	-1.19	*
		Cercis canadensis	1.5	1.21	-0.878	3.88	
		Quercus phellos	-0.829	0.745	-2.32	0.581	
		Ulmus americana	1.28	1.3	-1.21	3.86	
		Acer other	-2.83	1.14	-5.1	-0.673	*
		Prunus	1.42	0.988	-0.511	3.38	
		Quercus other	-2.13	0.867	-3.86	-0.517	*
		Other large	-0.486	0.612	-1.67	0.717	

		Other medium	-1.38	0.865	-3.09	0.244	
		Other small	-1.06	0.781	-2.63	0.48	
	Mulch	Correct	0.0539	0.16	-0.261	0.371	
		Volcano	0.179	0.323	-0.464	0.8	
	Landscaping	Minimal	0.798	0.267	0.273	1.3	*
		Moderate	0.588	0.382	-0.159	1.3	
		Extensive	0.451	0.397	-0.329	1.21	
	Urban tolerance		0.0431	0.489	-0.918	0.988	
	Site Type	Median	0.247	0.544	-0.814	1.32	
		Other Maintained	0.619	0.453	-0.265	1.53	
		Sidewalk Planting Strip	0.691	0.341	0.0439	1.38	*
		Yard	0.312	0.368	-0.384	1.06	
	Land Use	Commercial and Mixed	0.00348	0.362	-0.708	0.705	
		Institutional	0.694	0.28	0.142	1.24	*
		Natural area or Vacant	0.0918	0.335	-0.571	0.754	
		Other	0.049	0.486	-0.927	0.98	
	Basal area x Species Group	*Acer rubrum*	-0.414	0.15	-0.717	-0.119	*
		Cercis canadensis	0.0845	0.178	-0.263	0.439	
		Quercus phellos	-0.131	0.225	-0.58	0.303	
		Ulmus americana	0.059	0.208	-0.339	0.472	
		Acer other	-0.449	0.18	-0.808	-0.101	*
		Prunus	0.0672	0.15	-0.218	0.362	
		Quercus other	-0.325	0.157	-0.648	-0.026	*
		Other large	0.00062	0.114	-0.22	0.226	
		Other medium	-0.163	0.17	-0.498	0.169	
		Other small	-0.123	0.13	-0.371	0.134	
	Urban Tolerance x Site Type	Median	1.63	0.843	-0.04	3.27	
		Other Maintained	0.759	0.836	-0.85	2.35	

		Sidewalk Planting Strip	-0.453	0.549	-1.54	0.62	
		Yard	-0.286	0.57	-1.45	0.795	
Discoloration	intercept		0.674	0.564	-0.43	1.77	
	Segment	2	0.484	0.498	-0.477	1.46	
		3	0.297	0.542	-0.758	1.39	
		4	0.482	0.562	-0.594	1.61	
		5	-0.335	0.66	-1.61	0.957	
		6	0.421	0.502	-0.551	1.43	
		7	0.355	0.496	-0.614	1.33	
		Other	-0.0547	0.484	-0.983	0.903	
		8	0.754	0.532	-0.29	1.81	
		9	0.768	0.579	-0.363	1.94	
		10	0.184	0.513	-0.806	1.19	
		11	-0.307	0.577	-1.43	0.832	
		12	0.809	0.461	-0.0843	1.72	
		13	1	0.58	-0.103	2.15	
		14	-0.117	0.501	-1.11	0.879	
	Year	Year2023	0.508	0.114	0.282	0.734	*
	Basal Area		0.077	0.0816	-0.0854	0.235	
	Species Group	*Acer rubrum*	-0.113	0.56	-1.21	0.987	
		Cercis canadensis	0.781	1.11	-1.43	2.89	
		Quercus phellos	-0.265	0.499	-1.26	0.688	
		Ulmus americana	-0.277	1.12	-2.55	1.87	
		Acer other	0.0355	0.675	-1.3	1.39	
		Prunus	-1.34	0.915	-3.15	0.425	
		Quercus other	-1.32	0.553	-2.42	-0.233	*
		Other large	0.14	0.47	-0.783	1.06	
		Other medium	0.646	0.583	-0.492	1.8	
		Other small	-0.474	0.549	-1.55	0.612	
	Mulch	Correct	0.0407	0.132	-0.211	0.299	

		Volcano	-0.0779	0.259	-0.594	0.427	
	Landscaping	Minimal	-0.254	0.218	-0.69	0.17	
		Moderate	-0.452	0.297	-1.04	0.127	
		Extensive	-0.0487	0.301	-0.642	0.53	
	Urban tolerance		0.0762	0.316	-0.546	0.688	
	Site Type	Median	0.818	0.377	0.091	1.56	*
		Other Maintained	0.307	0.299	-0.286	0.901	
		Sidewalk Planting Strip	0.373	0.249	-0.115	0.852	
		Yard	0.281	0.282	-0.278	0.832	
	Land Use	Commercial and Mixed	-0.251	0.297	-0.844	0.334	
		Institutional	-0.252	0.243	-0.725	0.217	
		Natural area or Vacant	-0.269	0.307	-0.867	0.337	
		Other	-0.232	0.435	-1.11	0.601	
	Basal area x Species Group	*Acer rubrum*	-0.198	0.123	-0.435	0.0463	
		Cercis canadensis	-0.265	0.163	-0.596	0.0466	
		Quercus phellos	-0.524	0.167	-0.847	-0.198	*
		Ulmus americana	-0.328	0.183	-0.699	0.0149	
		Acer other	-0.0756	0.119	-0.309	0.162	
		Prunus	-0.315	0.137	-0.586	-0.0429	*
		Quercus other	-0.397	0.117	-0.637	-0.168	*
		Other large	-0.161	0.0954	-0.346	0.0258	
		Other medium	-0.0715	0.13	-0.329	0.18	
		Other small	-0.307	0.1	-0.507	-0.11	*
	Urban Tolerance x Site Type	Median	-0.356	0.548	-1.43	0.741	
		Other Maintained	-0.289	0.573	-1.4	0.817	
		Sidewalk Planting Strip	-0.289	0.365	-1.03	0.424	
		Yard	-0.409	0.392	-1.17	0.368	

Dieback	intercept		1.51	0.98	-0.38	3.44	
	Segment	2	-1.46	0.881	-3.21	0.316	
		3	-1.05	0.933	-2.86	0.736	
		4	-0.82	1.01	-2.84	1.13	
		5	-3.03	1.29	-5.68	-0.606	*
		6	-0.718	0.939	-2.61	1.05	
		7	-1.56	0.924	-3.44	0.223	
		Other	-0.893	0.87	-2.63	0.78	
		8	-0.167	0.986	-2.12	1.75	
		9	0.257	1.09	-1.9	2.37	
		10	-2	0.938	-3.89	-0.199	*
		11	-1.85	1.06	-3.98	0.191	
		12	-0.0766	0.825	-1.72	1.59	
		13	0.511	0.978	-1.41	2.38	
		14	-1.38	0.877	-3.18	0.269	
	Year	Year2023	0.204	0.137	-0.0625	0.47	
	Basal Area		0.0607	0.151	-0.233	0.357	
	Species Group	*Acer rubrum*	1.29	1.01	-0.631	3.34	
		Cercis canadensis	1.94	1.95	-1.85	5.77	
		Quercus phellos	2.12	0.883	0.462	3.92	*
		Ulmus americana	-3.14	1.9	-6.74	0.57	
		Acer other	0.545	1.24	-1.88	2.98	
		Prunus	2.73	1.63	-0.363	6.02	
		Quercus other	1.51	0.851	-0.113	3.26	
		Other large	1.07	0.789	-0.48	2.63	
		Other medium	0.103	0.998	-1.85	2.04	
		Other small	0.261	0.947	-1.58	2.11	
	Mulch	Correct	-0.122	0.202	-0.506	0.279	
		Volcano	0.625	0.506	-0.332	1.66	
	Landscaping	Minimal	-0.172	0.3	-0.771	0.423	
		Moderate	0.354	0.505	-0.628	1.35	

		Extensive	-0.818	0.49	-1.78	0.157	
	Urban tolerance		-0.07	0.615	-1.26	1.12	
	Site Type	Median	-0.13	0.683	-1.46	1.19	
		Other Maintained	-0.22	0.6	-1.37	1	
		Sidewalk Planting Strip	0.925	0.472	0.0116	1.87	*
		Yard	0.618	0.499	-0.371	1.58	
	Land Use	Commercial and Mixed	-0.219	0.555	-1.29	0.87	
		Institutional	-0.254	0.459	-1.15	0.654	
		Natural area or Vacant	-0.0527	0.602	-1.25	1.12	
		Other	-1.98	0.808	-3.57	-0.435	*
	Basal area x Species Group	*Acer rubrum*	-0.0365	0.221	-0.479	0.422	
		Cercis canadensis	0.0537	0.286	-0.502	0.608	
		Quercus phellos	0.638	0.33	0.0161	1.31	*
		Ulmus americana	-0.554	0.308	-1.15	0.0492	
		Acer other	0.0532	0.226	-0.401	0.491	
		Prunus	0.434	0.251	-0.0627	0.938	
		Quercus other	0.353	0.205	-0.0283	0.785	
		Other large	0.0471	0.169	-0.285	0.38	
		Other medium	-0.0749	0.219	-0.496	0.345	
		Other small	-0.17	0.179	-0.518	0.172	
	Urban Tolerance x Site Type	Median	-1.42	1.05	-3.5	0.577	
		Other Maintained	0.141	1.18	-2.14	2.43	
		Sidewalk Planting Strip	-1.2	0.719	-2.58	0.228	
		Yard	-0.334	0.712	-1.79	1.02	

Table 18: All parameter estimates for the univariate (crown stress) growing condition model in Durham. The baseline for street segment is the wealthiest street segment with at least 10 trees (arbitrarily numbered 1), the baseline for year is 2022, the baseline for Genus is all the groups with fewer than 40 observations (see Appendix B Table 4), the baseline for landscaping intensity is no landscaping, and the baseline for land use is residential. SE is the standard error, CI_025 and CI_975 together describe the 95% confidence interval around the parameter estimate. Sig95 indicates whether or not the 95% confidence interval crosses zero.

Predictor	Predictor Level	Estimate	SE	CI_025	CI_975	sig95
(Intercept)		0.004498	0.4857	-0.9407	0.9604	
Segment	2	-1.208	0.4686	-2.128	-0.2904	*
	3	-0.7692	0.4787	-1.71	0.1788	
	4	-1.062	0.5017	-2.043	-0.07549	*
	5	-1.142	0.5831	-2.297	-0.00478	*
	6	-0.5291	0.4759	-1.475	0.4008	
	7	-1.151	0.4584	-2.053	-0.2479	*
	Other	-0.9727	0.4425	-1.849	-0.1084	*
	8	-0.4283	0.4804	-1.382	0.5122	
	9	-0.222	0.5469	-1.297	0.8408	
	10	-1.167	0.4611	-2.075	-0.2636	*
	11	-1.447	0.5103	-2.452	-0.4538	*
	12	-0.9557	0.4236	-1.792	-0.1226	*
	13	-0.08216	0.5222	-1.099	0.9432	
	14	-1.833	0.4596	-2.735	-0.9383	*
Year	2023	0.2431	0.1062	0.03594	0.4498	*
BasalArea		-0.08432	0.03216	-0.1467	-0.02056	*
Species Group	*Acer rubrum*	1.297	0.2796	0.7529	1.846	*
	Cercis canadensis	2.111	0.3163	1.502	2.745	*
	Quercus phellos	1.487	0.3118	0.8788	2.098	*
	Ulmus americana	1.408	0.3025	0.8195	2.004	*
	Acer other	0.7174	0.2784	0.1635	1.262	*
	Prunus	0.7904	0.2746	0.2527	1.327	*
	Quercus other	0.9417	0.2946	0.3716	1.515	*
	Other large	1.372	0.2526	0.8819	1.866	*
	Other medium	1.02	0.2685	0.4948	1.55	*
	Other small	1.466	0.2435	0.991	1.946	*

Powerline		-0.3617	0.139	-0.6354	-0.08837	*
Mulch	Correctly	-0.2208	0.1194	-0.4531	0.01363	
	Volcano	-0.3469	0.2358	-0.8084	0.1217	
Landscaping	Minimal	-0.378	0.1926	-0.7512	0.003994	
	Moderate	-0.4199	0.2712	-0.9514	0.1077	
	Extensive	-0.3566	0.2713	-0.8828	0.1772	
Urban tolerant		-0.304	0.2932	-0.8814	0.2763	
Site type	Median	0.6785	0.3516	-0.01713	1.361	
	Other Maintained	0.3502	0.2712	-0.1831	0.8806	
	Sidewalk Planting Strip	0.5024	0.2269	0.05451	0.9463	*
	Yard	0.299	0.2505	-0.1874	0.7895	
Land use	Commercial or Mixed	-0.3199	0.2553	-0.8235	0.1769	
	Institutional	-0.3106	0.2151	-0.7304	0.1091	
	Natural area or Vacant	-0.092	0.2571	-0.598	0.4076	
	Other	-0.487	0.3741	-1.212	0.248	
Urban tolerance x Site type	Median	-1.405	0.5099	-2.417	-0.418	*
	Other Maintained	0.1612	0.4902	-0.8017	1.133	
	Sidewalk Planting Strip	-0.4768	0.3277	-1.119	0.1547	
	Yard	0.03343	0.34	-0.6438	0.6937	

Table 19: Residual covariance between tree stress variables in Chicago. ± values are the standard errors. Since the tree stress variables (defoliation, discoloration, dieback) were modeled together in GJAM, the output includes a residual covariance matrix that describes the correlation between tree stress variables after the predictors have been accounted for. The matrix is symmetrical so only one set of pairwise comparisons are shown.

Tree Stress Metric	**Defoliation**	**Discoloration**	**Dieback**
Defoliation	1	0.2759 ± 0.0987	-0.0708 ± 0.1320
Discoloration		1	0.1787 ± 0.0809
Dieback			1

Table 20: Residual covariance between tree stress variables in Durham. ± values are the standard errors.Since the tree stress variables (defoliation, discoloration, dieback) were modeled together in GJAM, the output includes a residual covariance matrix that describes the correlation between tree stress variables after the predictors have been accounted for. The matrix is symmetrical so only one set of pairwise comparisons are shown.

Tree Stress Metric	Defoliation	Discoloration	Dieback
Defoliation	1	0.0471 ± 0.0772	0.0956 ± 0.1135
Discoloration		1	-0.0165 ± 0.1020
Dieback			1

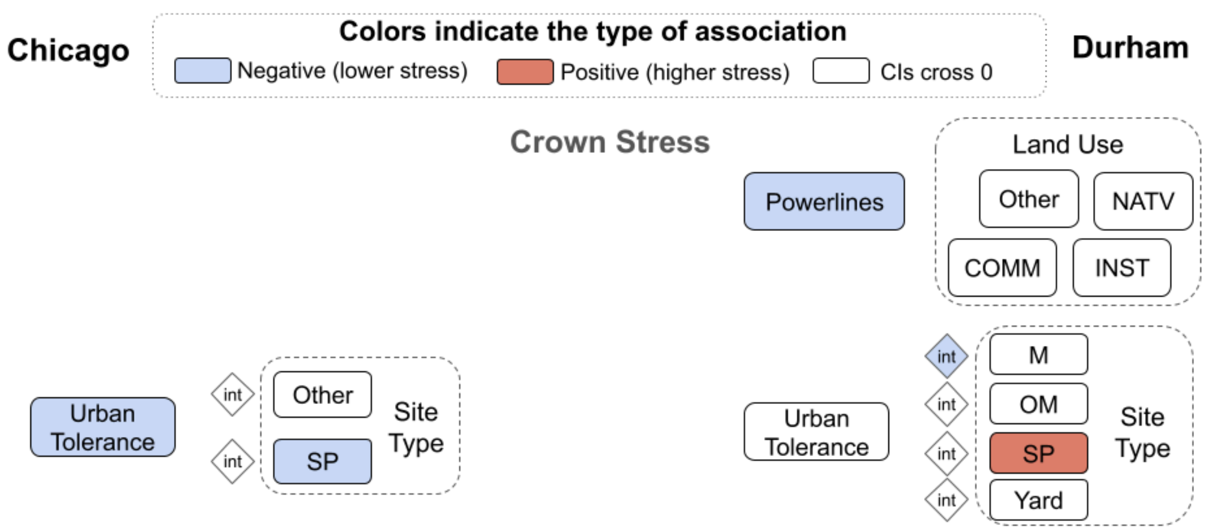

Figure 4: Relationships between site condition predictors variables and crown stress in Chicago and Durham. Box colors signify the type of association: negative (light blue), positive (red), or with confidence intervals that cross zero (white). The abbreviations for land use are: commercial and mixed-use (COMM), managed park (MP), institutional (INST), multi-family residential (MFR), and natural area/vacant lot (NATV). The abbreviations or site types are sidewalk planting strip (SP), median (M), and other maintained area (OM). The baseline category for site type is sidewalk cutout for Chicago and sidewalk cutout and other hardscape for Durham. The baseline category for land use is residential for Durham. Full parameter estimates with standard errors and confidence intervals can be found in supplemental Tables 15 and 17.

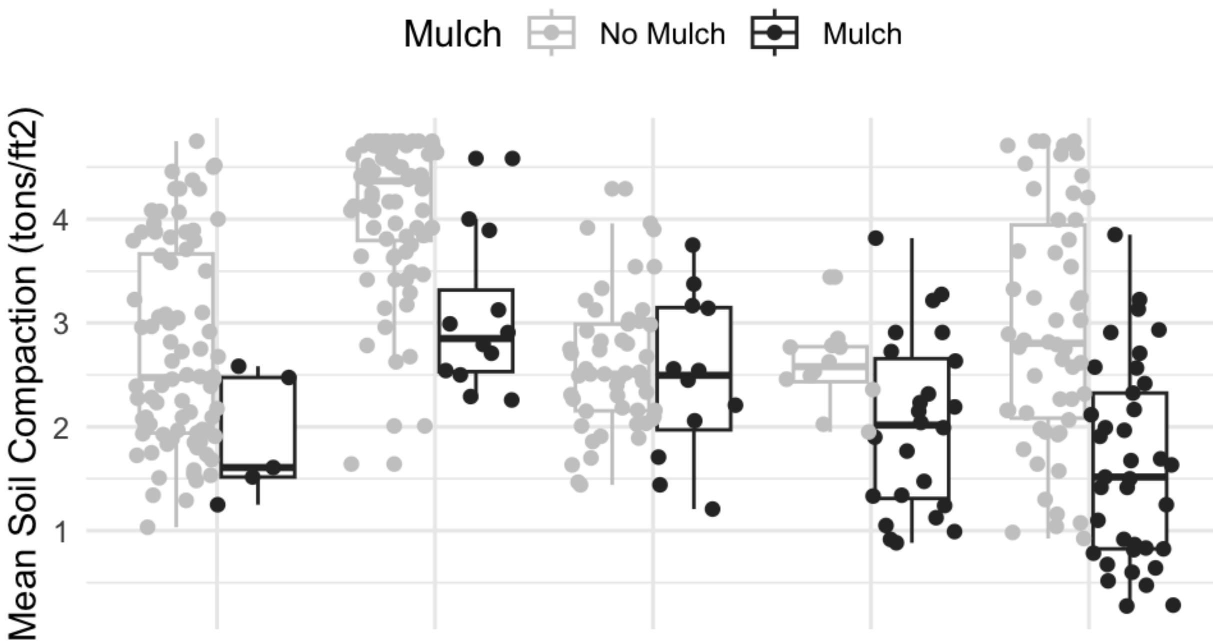

Figure 5: Pilot soil compaction data and presence of mulch for a subset of Chicago segments in 2022 (n=367 trees). Each pair of boxplots (grey and black) refers to a different pilot street segment in Chicago. Soil compaction was measured with a pocket penetrometer at 6 points around the base of the trunk, within the canopy drip line. The presence or absence of mulch was noted as part of field data collection.

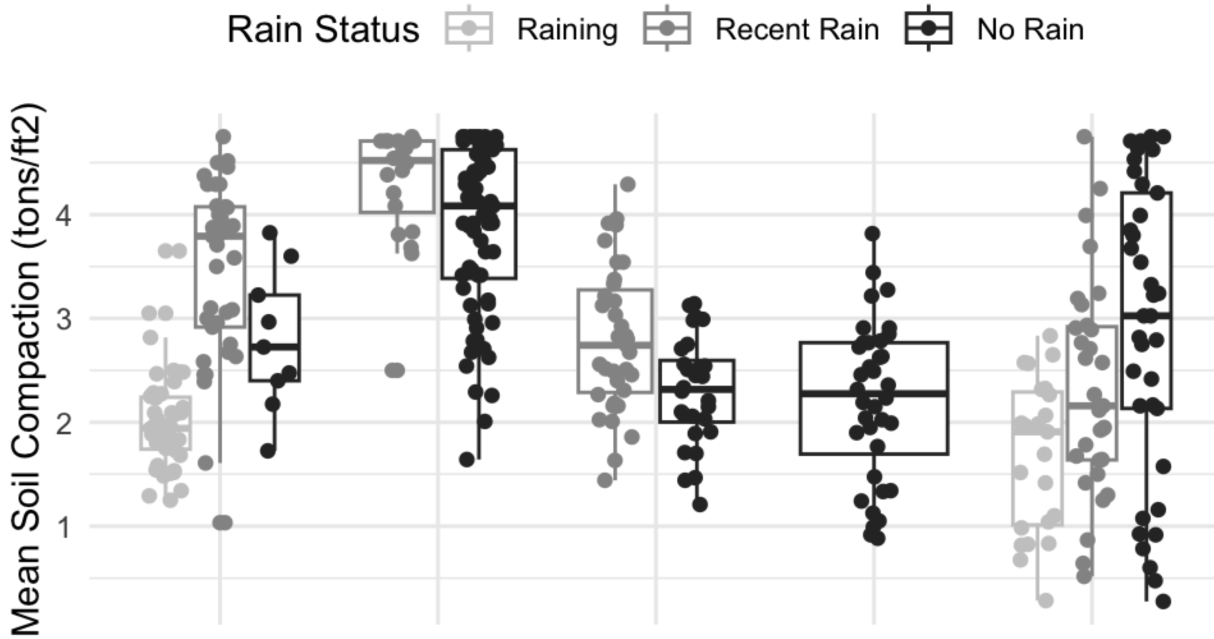

Figure 6: Pilot soil compaction data and whether fieldwork occurred in the rain for a subset of Chicago segments in 2022 (n=367 trees). Each trio or duo of boxplots (light grey, grey, black) is from a different pilot street segment in Chicago. Soil compaction was measured with a pocket penetrometer at 6 points around the base of the trunk, within the canopy drip line. Days with recent rain were those where at least 2.5cm of rain was recorded in the previous 24 hours.